ITA0610 - Machine Learning

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Experiment: Candidate Elimination Algorithm

Aim

To implement and demonstrate the **Candidate Elimination Algorithm** in Python using a **CSV file** and obtain the set of all hypotheses that are consistent with the given training examples.

Algorithm

1. Import the required Python libraries.
2. Read the training dataset from a CSV file.
3. Initialize the **Specific Hypothesis (S)** with the first positive training example.
4. Initialize the **General Hypothesis (G)** with the most general hypothesis.
5. For each training example:
   - If the example is **Positive (Yes)**:
     - Update the Specific Hypothesis by generalizing it where necessary.
     - Remove inconsistent hypotheses from the General Hypothesis.
   - If the example is **Negative (No)**:
     - Specialize the General Hypothesis to exclude the negative example.
6. Repeat until all training examples are processed.

7. Display the final Specific Hypothesis (S) and General Hypothesis (G)

PROGRAM:

```
import pandas as pd
import numpy as np

data = pd.read_excel("trainingdata.xlsx")

print("Training Data:\n")
print(data)

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

def learn(concepts, target):

    specific_h = concepts[0].copy()

    general_h = [["?" for i in range(len(specific_h))]
                 for j in range(len(specific_h))]

    for i, h in enumerate(concepts):

        if target[i] == "Yes":
```

```
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = "?"
                    general_h[x][x] = "?"

        elif target[i] == "No":

            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
                else:
                    general_h[x][x] = "?"

    general_h = [g for g in general_h if g != ["?"] * len(specific_h)]

    return specific_h, general_h

specific, general = learn(concepts, target)

print("\nFinal Specific Hypothesis:")
print(specific)

print("\nFinal General Hypothesis:")
for g in general:
    print(g)
```

OUTPUT:

```
print(specific)

print("\nFinal General Hypothesis:")
for g in general:
    print(g)
```

```
Training Data:

     Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool   Change        Yes

Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Result

The **Candidate Elimination Algorithm** was successfully implemented and executed using the given training dataset. The algorithm generated the **Specific Hypothesis (S)** and **General Hypothesis (G)**, representing the set of all hypotheses consistent with the training examples. The final specific hypothesis obtained was:

['Sunny' 'Warm' '?' 'Strong' '?' '?']

The final general hypotheses obtained were:

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

Hence, the Candidate Elimination algorithm successfully identified the version space consistent with the given training data.